Gebrüder Weiss CRO team

INCHEON RoboCup

Gebrüder Weiss

RoboCup Junior   Rescue   HROBOS

Our team

David Pongrac

Gregor Klarić

Team mentors

Marko Pongrac & Jelka Hrnjić

David Pongrac

Team captain, programmer

"Being the leader of the team, I made all software parts of our robot, from mapping and navigation to victim detection."

other competitions:

Croatia National 2023,
Rescue simulation, 1. place
Croatia National 2024,
Rescue simulation, 2. place
World RCJ 2023.,
Rescue simulation, participated
World RCJ 2024.,
Rescue simulation, Team spirit award

Gregor Klarić

Electronic & Mechanic designer

"Using the knowledge I've gained in electronics and mechanics, I made decision about design of the robot."

other competitions:

Croatia National 2024,
Soccer open, 1. place
Slovakia Open 2024,
Soccer open, 1. place
World RCJ 2024.,
Soccer open, 4. place
European RCJ 2025.,
Soccer Lightweight Entry, 2. place

Our team's history of participation and awards:

Croatia National 2022, Rescue maze Entry, 3. place
Austria Open 2022., Rescue maze Entry, 1. place
European RCJ 2022., Rescue maze Entry, participated
Asia-Pacific 2024., Rescue maze, 2. place
Croatia National 2023., Rescue maze Entry, 1. place
European RCJ 2023, Rescue maze Entry, participated
Croatia National 2024, Rescue maze, 2. place
Slovakia Open 2024, Rescue maze, 1. place
Austria Open 2024, Rescue maze, 3. place
European RCJ 2024, Rescue maze, participated
Croatia National 2025, Rescue maze, 2. place
Austria Open 2025, Rescue maze, 2. place
European RCJ 2025, Rescue maze, participated (4. place)
World RCJ, Salvador 2025, Rescue maze, participated
Croatia National 2026, Rescue maze, 1. place
Austria Open 2026, Rescue maze, 2. place

HDR

We are robotics enthusiasts from Zagreb for over ten years. We practice robotics at the workshops of the Croatian Robotics Society.

Croatia, a country so small, yet its people so powerful.

Software

Before we constructed our robot, we made the mapping system. While making it, we began working on the its system for victims. Those 2 were able to be developed independently from each other.

Victim detection

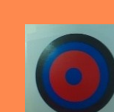
Cameras – our tools for the detection

Our robot has on RPI Fisheye 160^{0} camera on each of its sides. They allow him to see almost 16-20 cm of wall space, about 60% the size of one tile.

Our weapon for victim detection – YOLO algorithm

For the victim detection, our first layer is YOLOv26 algorithm, which allows us to know whether the victim is Ψ, Φ, Ω or cognitive (will be latter processed). We collected them through the year and now have more then 2000 images, labelling them and splitting into 50-50 split before feeding them into the model. We also took pictures of various fake victims (for example, coloured letters) and made our model not detect them. Due to our experience from last year, we trained fewer models while achieving nearly 0% misidentification rate.

The second layer is processing of cognitive victims. Using several algorithms (Gaussian blur, Canny algorithm, Floodfill) we segment the cognitive victims in order to determine the class and weight of each circle. We than multiply the value of each color by the width of a circle to get its value.



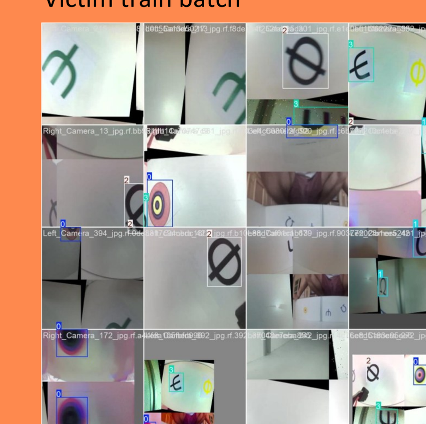
region of interest

combined edges raw

Floodfill from midpoints

normalized confusion matrix

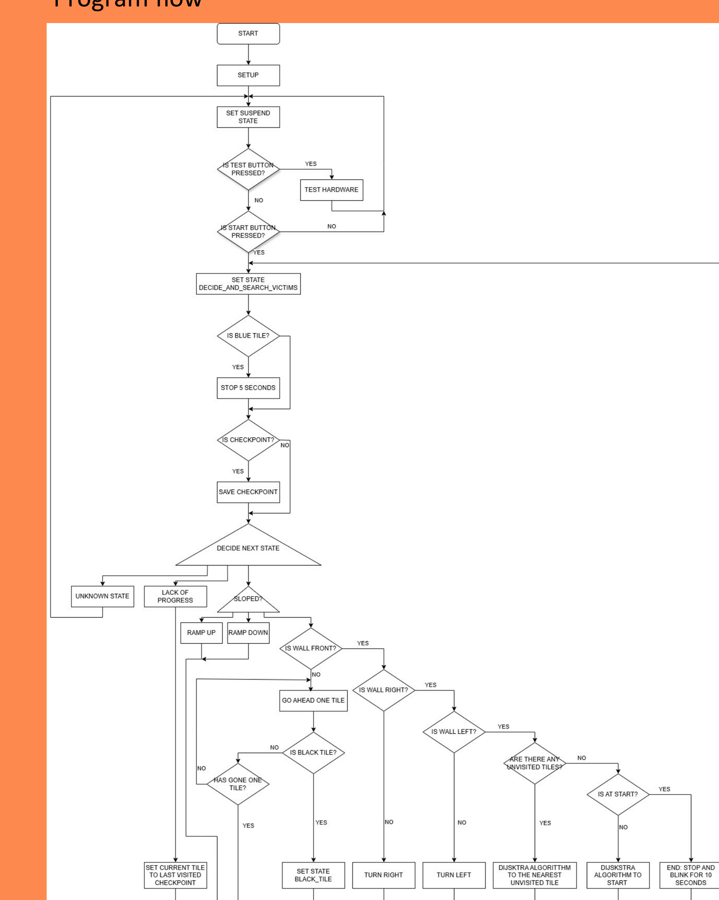
Victim train batch



Mapping of victims

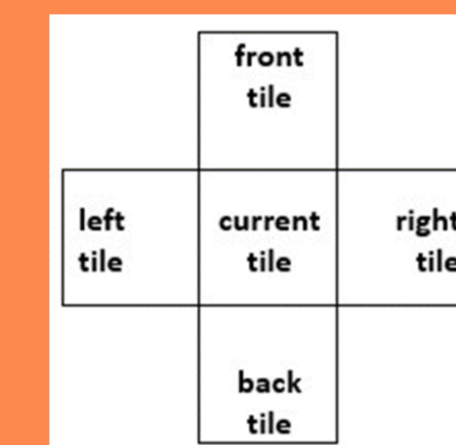
Each tile object also has an array that contains the information about the type of victim on the wall (absence of victim is said as NONE). With this system, the robot doesn't waste rescue kits when it's found again.

Program flow



Mapping and navigation system

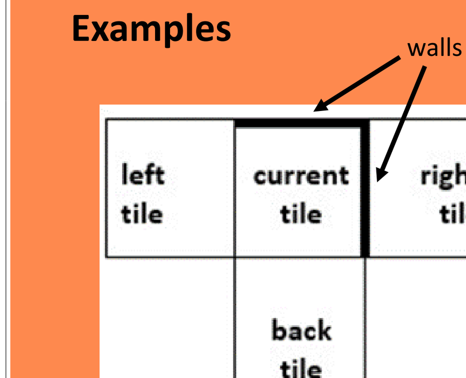
Our navigation and mapping system and very intertwined. It works as following:



(front tile, right tile, back tile, left tile)

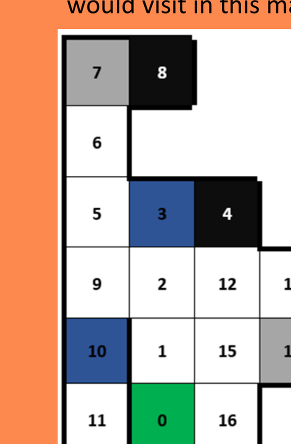
1. Tile current visited was created as an unknown tile in relationship to the last visited tile.
2. Give the newly explored tile its tile type: blue, black, silver, red, ramp or stairs.
3. Check the surrounding walls:
   - if there is a wall mark as a nullptr
   - if there is no wall check the X, Y and Z difference
   - if any of the neighbouring tiles was already explored, connect them together using pointers.
   - For the remaining tiles that are both unexplored and unmapped, create new tile objects as unknown and connect them to the current tile using pointers, we are going to explore them latter.
4. Decide which tile we are going to explore next:
   - Firstly, check neighbouring tiles if we can visit them in order front → right → left.
   - If all surrounding tiles are walled off or they were already explored, use the Dijkstra's algorithm to find the latest tile we labelled as unknown (last added to stack).
   - If there are no unexplored tiles and the robot is not already on the starting tile, use the Dijkstra's algorithm to return back to it.
   - If the robot is on the start tile, initiate the end protocol: stop for 10 seconds.

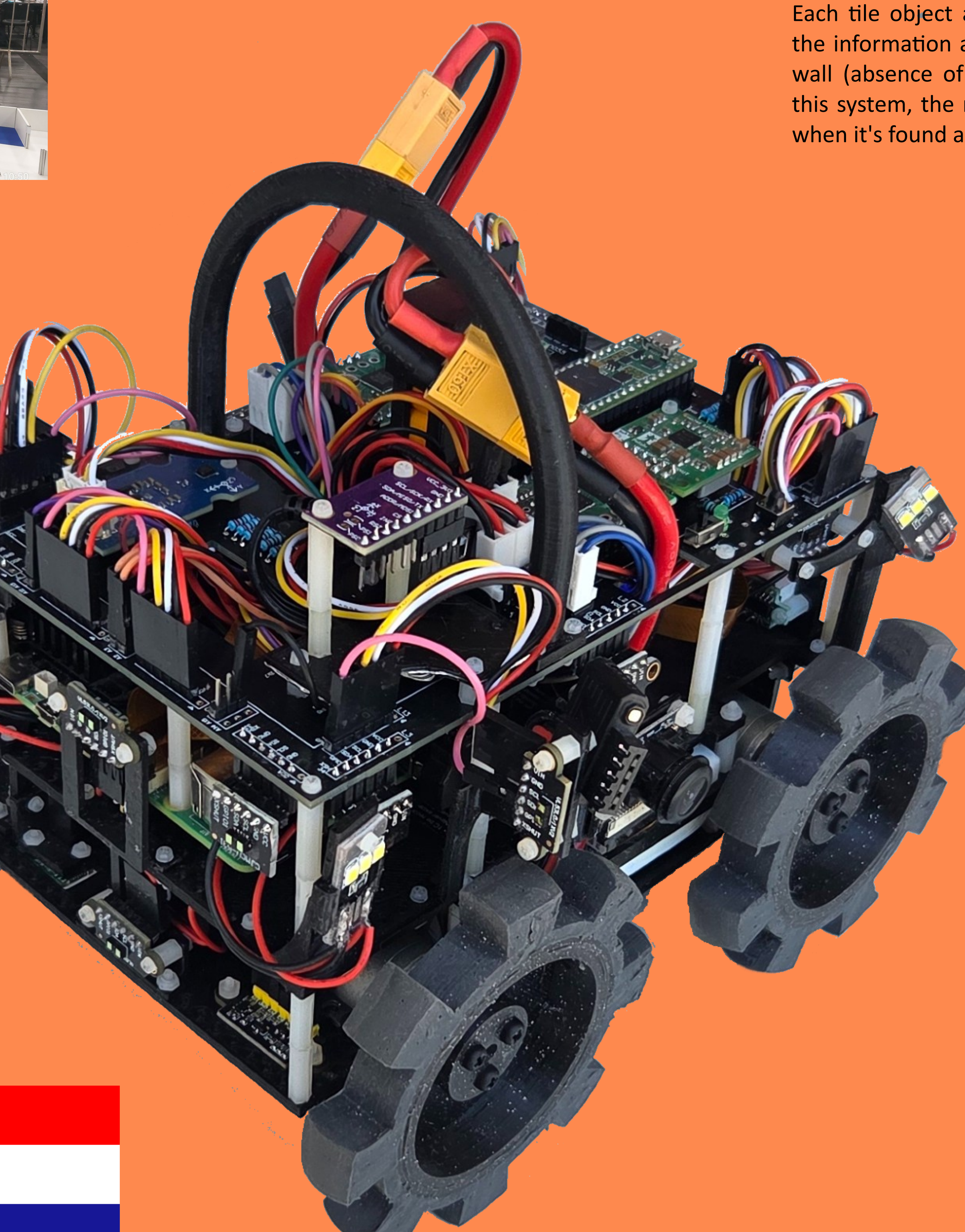
Examples



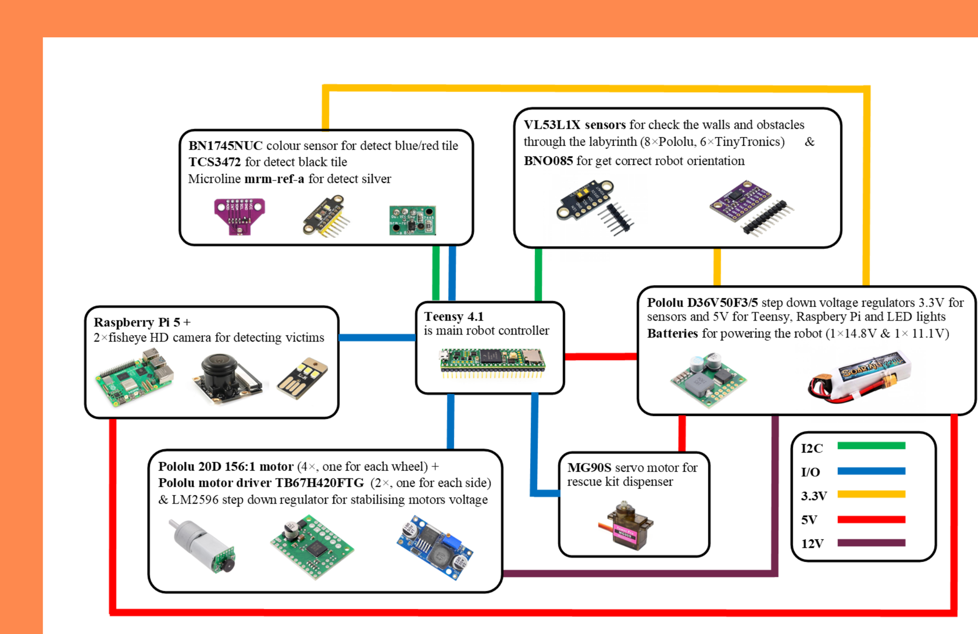
(nullptr, nullptr, back tile, left tile)

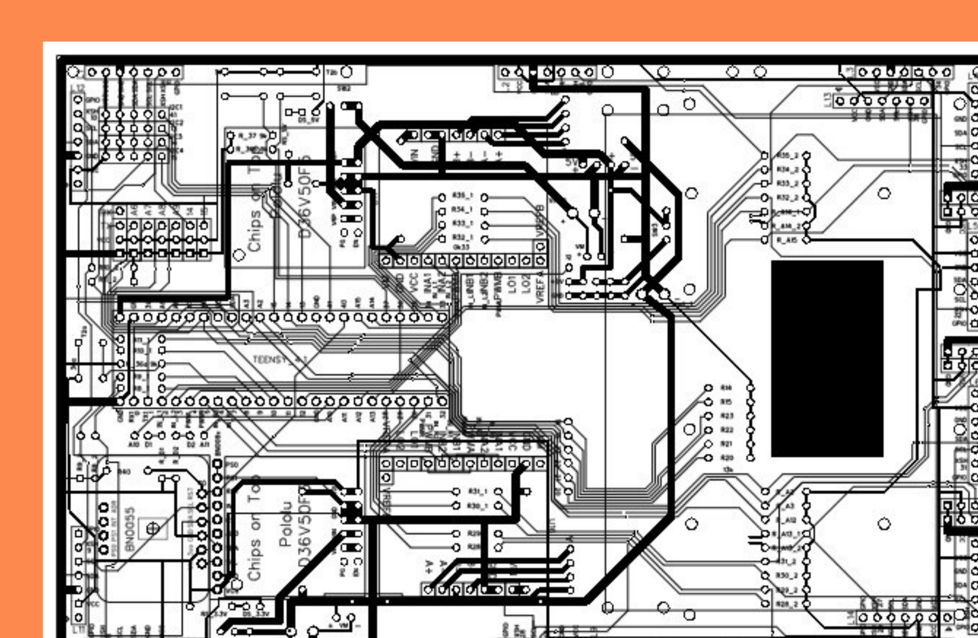
Order of tiles our robot would visit in this maze





Hardware





Our robot was continuously developed since RCAP 2023 Asia-Pacific in Pyeongchang. From the very beginning it had a rear suspension for the easier pass over speedbumps. It also had 12 lidars and compass for the movement in the arena.

Since 2024 the robot had a PCB which we made by ourselves. On it we connected all sensors, electronic compartment and the ***brain of our robot – Teensy 4.1.*** For the coloured tiles we tried various combinations. In the end, we decided to use BN1745NUC colour sensor placed in the middle of the robot's lowest layer to minimise the effect of outside lighting. After we had problems with the detection of the black tile, we decided to replace one of IR sensors with VEML6040 colour sensor because of dimensions similar to previous IR sensor, while the other IR sensor we still use for the detection of checkpoint.

In order to secure the consistent behaviour of the robot we use special voltage stabilisers for the motors that have a defined input voltage of 11 V because of battery characteristics. RPI 5 is a high energy consumer. It is, as well as other 5V components powered by a 14.8V LiPo battery, while the other components use 11.1V LiPo battery.

During the year 2025 mayor changes were made in the design of the wheel to facilitate the crossing over 2 cm speedbumps, mostly on the ramps. Kit dispenser was also redesigned, since the old one had a problem of the rescue kits often got stuck due to which servo was damaged. Sensors for the detection of obstacles were also replaced with lidars, and position of other lidars were corrected for better detection inside the maze.

During the last year mayor changes was made in the position of the compass, according to the which we have with it in Salvador, we try few different models. At the end we use the old one, but on the new position upper on the front side. Also we decrease rescue kit dispenser because we need place for only 8 rescue kits. Last thing which we are change was colour sensor for detect black tile, wher we use TCS3472 instead VEML6040 according better performance of the new one.